A. Describe the problems Company A has with network security and infrastructure using details from one or more of the "Company A Organizational Chart," "Company A Risk Analysis," and "Company A Visio Diagram" supporting documents.

Company A's network security and infrastructure are afflicted by several problems that are highlighted in the "Company A Risk Analysis" (n.d.) document. The company's network consists of several servers including an FTP server, data storage cluster server, web server, exchange server, and application server. Each of the servers runs Windows Server 2008-2012. In addition, there are approximately 86 workstations running Windows 10 Pro and 6 are configured for remote desktop ports 88-93. There are 14 Dell Inspiron laptops running Windows 7 and 6 laptops running Windows 10 Pro (20 laptops total). There are 4 Cisco 2960 48/4 and 3 Cisco 2960 24/4 switches, a Cisco PIX 515E firewall, and a Cisco 2811 router. The cabling used in the network is Cat5e. A thorough review of the "Company A Risk Analysis" (n.d.) and "Company A Visio Diagram" (n.d.) documents demonstrate the vulnerabilities in the network's current setup.

Risk #1: Remote desktop at open ports 88-93. Although the technical department must have a way to remotely access workstations and employees working from home need access to the network, utilizing open ports is unsafe. The open ports pose a serious security threat, as attackers can attempt to access these ports to gain access to the entire network. For example, attackers can access these ports and use networking capturing tools to gain sensitive information that employees are sending and/or receiving over the internet. More secure alternatives include using a Virtual Private Network (VPN) and Remote Desktop Software that have built-in security

features like multi-factor authentication.

Risk #2: Outdated Windows Server 2008 and 2012. Both of the servers that Company A is currently using have reached their end lives. Microsoft ended its support for Windows Server 2008 on 01/14/2020 and will end its support for Windows Server 2012 this year in October (dknappettmsft & v-alje, 2022). This is concerning because the company has been using a server for the past 3 years that has not received any security updates or patches to the operating system for Microsoft. The Windows 2012 server is quickly approaching its end of life and both servers need to be updated for more security.

Risk #3: Laptops running Windows 7. According to Microsoft, support for Windows 7 ended on January 14, 2020 (Microsoft, n.d.). The 6 laptops that are still using Windows 7 should be upgraded to Windows 10 Pro for the same reasons that the servers need to be upgraded.

Risk #4: Cisco PIX 515E firewall and Cisco 2811 router. The Cisco firewall and router are both obsolete products with no support. The firewall's end of life was 05/25/2023 (Cisco, 2017) and the router reached its end of life on 11/01/2016 (Cisco, 2019). As a security professional, it is alarming that Company A has been using such outdated equipment for almost 10 years. Due to how antiquated the firewall is, attackers can easily gain access to the network and employees are highly susceptible to downloading and accessing malicious materials over the network. In the merged network, the firewall and router should be upgraded.

Risk #5: Full access privileges granted to every employee. Currently, all employees are granted access to all systems except the payroll system. As a property and casualty insurance underwriting company, Company A has highly sensitive information.

Insurance underwriters store customers' personal information, as well as valuable proprietary information related to the pricing of policies. The lack of network segmentation in the current setup makes it easier for employees to access information they shouldn't and/or don't need access to. Segmenting the network based on departments or users' needs will make it more secure and increase network speeds.

Risk #6: Password changes not enforced. At this time, employees are not required to change their passwords at Company A. To minimize the need to remember numerous passwords, individuals tend to reuse passwords for multiple resources. Passwords can also be written down in places where other people who should not have access will see them. Password changes should be enforced to prevent the reuse of passwords and tighten the security of the network. Also, password cracking is a primary way that cyber criminals and other unauthorized parties gain access to networks and sensitive data. If passwords are never changed, it's easier to crack them. The company has legal and moral obligations to protect customer data, which they are not doing right now.

Risk #7: IT Department structure. From the "Company A Visio Diagram" (n.d.) document, it is evident that there is only 1 network administrator and 1 help desk PC technician. The organization doesn't have any personnel who specialize in information security. Upon merging, the IT department should be restructured to include at least 1 person who specialized in cybersecurity. Cybersecurity professionals are experts in information security when it comes to best practices and advancements in technology. Hiring a cybersecurity professional will assist with making the new network more secure by handling incident management, identifying vulnerabilities, threats, and the best way(s) to mitigate them.

Risk #8: User accounts are no longer removed. The "Company A Risk Analysis" (n.d.) document highlights that employees who may not work at Company A anymore still have access to the network. This is a major security concern because a disgruntled ex-employee can access proprietary data with malicious intent.

B. Run Zenmap and OpenVAS on Company B's network(s) using the "Performance Assessment Lab Area" web link: See attachments.

C. Describe the potential impact of two network security or infrastructure problems of Company B using details from the outputs of Zenmap and OpenVas, including rationale for *each* problem.

1. **Explain how correcting each of these problems will improve the functioning of the merged network.**

Company B has several security vulnerabilities that were highlighted in Zenmap and OpenVas scans completed from the Kali machines. Scanning the Ubuntu router from the Kali 1 machine revealed several security risks associated with user xprosp3 (IP address 10.10.1.2) and user student-pc (IP address 10.10.1.3).

Risk #1: Devices Running Windows XP and Windows 7. Nmap and OpenVas details regarding user xprosp3 showed that the device is running Windows XP. Windows XP reached its end of life on 04/14/2009 and Microsoft ended support for the product on 04/08/2014 (Microsoft, n.d.) Since that date, the operating system has not received any updates or security patches. The operating system on user student-pc is Windows 7 which is another outdated operating system. Updating the operating systems on these devices will make the network more secure. Windows 10 will be supported until 10/25/2025 (Microsoft, n.d.), so the merged organization can save money in the short run by upgrading to Windows 10 vs Windows 11. Windows 10 has built-in security

features that will make the network more secure and improve protection against malware, unauthorized access, and data breaches.

Risk #2: Open ports 135, 139, and 445. An analysis of the scans showed that ports 135, 139, and 445 are open on both devices. These ports used for remote access and file sharing are known to be commonly used in security attacks. They have a history of being used in attacks that allowed the spread of malware in a network or allowed unauthorized access to a network. Considering that the merged network will contain sensitive proprietary information and customer data, ports 135 and 139 should be closed. However, Windows devices still use port 445 for printing, file sharing, and inter-network communication so closing this port may cause some services or programs not to work properly. To properly secure port 445, a firewall can be configured to block unauthorized traffic. Closing those ports reduces the visibility of the merged network's services to the outside world, making it harder for attackers to identify vulnerabilities or exploit weaknesses.

Risk #3: Message signing disabled on user student-pc (IP address 10.10.1.3). Message signing is an important security feature that verifies the authenticity of data communications in a network. It ensures that a message has not been tampered with by an attacker or other unauthorized party. Message signing can be enabled on the device by going into the device settings. This is a simple and free security patch that will be addressed in the merged network.

Risk #4: OPNsense firewall running FreeBSD 11.2. OPNsense is a free and reliable open-source firewall that is used by many businesses for their security needs. While there is nothing wrong with the firewall of choice, the operating system is outdated and needs to be updated immediately. The purpose of a firewall is to block

unwanted traffic and help prevent attacks on the network, but if the operating system is outdated then it may not function how it should. The firewall operating system can be upgraded to the latest FreeBSD version which is 23.1 (OPNsense, n.d.). OPNsense also offers a business support license for $339.30 per year which is a suggested purchase for technical support from the company instead of the user community (OPNsense, n.d.). However, since Company A will be using a Cisco 4331 router that comes with a built-in firewall and VPN, it will make more sense to purchase that router for Company B as well. In the merged network, both companies should have the same security features including using the same VPNs. Additionally, Cisco offers a warranty on their products along with 24/7 support.

Risk #5: Outdated servers. A review of Company B's network topology in the performance lab shows 3 servers: Windows 2012 server (IP address192.168.25.3), Windows 2016 server (IP address 192.168.25.2), and Windows 2008 server (IP address 192.168.25.4). All of these operating systems have surpassed or are quickly approaching the end of their life cycles. To improve the overall function of the network, they should all be upgraded to Windows Server 2022. It is the latest version of Windows server and it will continue to receive updates and security patches for the next 8 years. Servers with newer software will allow the IT department to use modern tools and protocols to protect the network against potential threats. In the financial services and insurance industries, compliance with specific regulations and standards is crucial. Upgrading to Windows Server 2022 helps ensure that our servers meet the latest compliance requirements, especially those that relate to consumer data protection.

D. Provide a network topology diagram using Microsoft Visio or a

similar software tool that represents the proposed merging requirements from the scenario of Company A and Company B, including remediation of *all* existing infrastructure problems described in part A and part C: See attachment.

Company A in Merged Network: A total of 8 risks were identified in Company A's network. Since all the risks cannot be physically represented in the merged network topology, it should be assumed that Company A risk #5 (full access privileges granted to every employee), risk #6 (password changes not enforced), risk #8 (IT department structure) and risk #9: user accounts are no longer removed were addressed. In the merged network, users are forced to do password changes periodically on the startup screen. In addition, Upon termination/resignation, user accounts are removed

Company B in Merged Network: A total of 5 risks were identified in Company B's network. Since all the risks cannot be physically represented in the merged network topology, it should be assumed that Company B risk #2 (Open ports 135, 139, and 445) and risk #3 Message signing disabled on user student-pc are addressed. In the merged network, the OPNsense firewall has been replaced with the firewall provided by the Cisco 4331 router and configured to block unauthorized traffic. Ports 135 and 139 are closed and message signing is enabled on user student-pc (IP address 10.10.1.3).

E. Identify which layer of the OSI model and layer of the TCP/IP protocol stack apply to *each* component in the merged network topology diagram.

Device	OSI Layer	TCP/IP Layer
Cisco 4331 Routers	Network	Network/Internet (Layer 2)

Cisco 2960-X switches	Data Link	Network Interface (Layer 1)
Cisco Firewall	Network & Transport	Network/Internet (Layer 2) & Transport (Layer 3)
Workstations	Session, Presentation & Application	Application (Layer 4)
Laptops (Remote Users)	Session, Presentation & Application	Application (Layer 4)
Email Server	Transport & Application	Transport (layer 3) & Application (Layer 4)
FTP Server	Application	Application (Layer 4)
Application Server	Presentation & Application	Application (Layer 4)
Windows 10 Operating System	Application	Application (Layer 4)
Cisco VPN	Network & Application	Network/Internet (Layer 2) & Application (Layer 4)
Printer	Presentation & Application	Application (Layer 4)
Cat5e Cabling (Not newly purchased)	Physical	Physical(Layer 1)

F. Justify the retention or deletion of four existing components as suggested in your proposed network topology diagram. Include how each component's retention or deletion and any newly required additions to the network address *both* security concerns and budgetary restrictions found in the scenario.

When designing the topology of the merged network, it was important to consider security concerns and budgetary restrictions. Through the careful analysis of the security implications and cost-effectiveness of each component, we were able to design a network that addresses the needs of our internal and external stakeholders. At

this time, the focus will be placed on four components that were deleted in the merged network.

Component #1 Deletion – Replaced all outdated servers at Company A and Company B with Windows 2022. The servers at both companies were outdated and posed a threat to the security of the merged network. The servers were past the time period that Microsoft would provide technical support or security updates. Upgrading the servers to Windows Server 2022 ensures that the network is supported by Microsoft and updates to patch security vulnerabilities are received. In addition, we will be in compliance with current regulatory guidelines. If an attack on the network occurred and the servers were not upgraded, the merged organization could be legally liable for failing to secure sensitive data. It will cost $8,552 to upgrade all servers to Windows Server 2022.

Although there is a significant upfront cost to upgrade all the servers, it will save the organization money in the long run. Microsoft will continue to provide support and security updates for Windows Server 2022 until 2026 (Microsoft, n.d.). Windows Server 2022 has built-in security features such as improved authentication protocols that the older versions lack (Robinharwood et al., 2018). By proactively upgrading the servers, the organization can save money by avoiding expensive incident response costs after an attack on the network.

Component #2 Deletion – Replaced routers at Company A and Company B with Cisco 4331 Routers. The router being used at Company B was an older model that posed a significant risk to the security of the new network. The prior router being used at Company A (Cisco 2811) was released almost 20 years ago in 2004 and does not meet today's requirements for a secure network. It was discontinued in

2016, Cisco no longer supports it and it will likely be incompatible with the components being added to the merged network.

It will cost $6,097.98 to replace the routers at both companies. Replacing the prior routers with Cisco 4331 offers several benefits including improved network speeds and overall performance of the network. The built-in firewall and VPN that the Cisco 4331 offers will save money immediately. There will be no need to purchase VPN or firewall services from a third party because these services are already integrated into the router for free. Replacing the router at Company B with the same router that Company A has, allows for easier network management. If both companies have the same router, firewall, and VPNs then the network configuration and security policies will be the same creating consistency. Routers control network traffic and help keep unauthorized users out. Keeping outdated routers to save money could compromise the network and cost the organization more than it would cost to replace them.

Component #3 Deletion – Upgraded all laptops and desktops to Windows 10. Laptops and desktops at both companies were running a wide range of outdated operating systems such as Windows 7, Windows 8, and Windows XP. All of those operating systems do not receive bug fixes or updates from Microsoft. When designing a secure network, it's important that devices are running up-to-date programs.

The approximate cost to upgrade the operating systems on the devices that had outdated systems will be $2,199.80. Windows 10 has important security features that its predecessors lack. The operating system has built-in security tools that can be used to mitigate threats to the network for free. For example, Windows 10 has a security feature called Device Guard that helps to protect a device from malware. This is a feature that

enhances security because an administrator can use it to ensure that devices only run trusted applications. The devices should also be upgraded to Windows 10 because modern applications are developed to work with a newer operating system like Windows 10. The organization will run into compatibility issues if the systems aren't updated.

Component #4 Retention – Cat5e cabling. The current cabling being used can be retained in the merged network without compromising network efficiency or security. Replacing the cables isn't necessary, as Cat5e cables have a maximum bandwidth of 1,000 Mbps. This is sufficient for the needs of the network and retaining the cables will save money. With a budget of $35,000, it was important to prioritize replacing devices and software that posed a direct threat to the security of the network. Replacing all of the cables at Company A and Company B's physical locations would have likely placed us over budget. The components in the merged network are compatible with Cat5e cables and the requirements of the network are being met without spending additional money on new cables.

Price List

Hardware/Software	**Price**	**Quantity**	**Total**
Cisco Catalyst 2960X 48 Port Switch	$1,542.59	6	$9,255.54
Cisco 4331 Routers (Built-in VPN and Firewall)	$3,048.99	2	$6,097.98
Windows Server 2022	$1,069.00	8	$8,552
Windows 10 Licenses	$109.99	20	$2,199.80

Budget: $35,000
Total Costs: $26,105.32 **Note: Links for all items purchased can be found in the citations list.*
Remaining funds: $8,897.68

G. Explain how **two** secure network design principles are included in your proposed merged network topology diagram.

The merged network was designed with secure network principles in mind such as defense in depth and least privilege (Chapple et al., 2021). Prior to the merger, each company had major vulnerabilities in their networks that needed to be corrected to mitigate the likelihood of attacks, especially malware like viruses, trojans, and worms (Panko & Panko, 2019).

Principal #1 — Defense in depth. This principle emphasizes the importance of a layered approach to network security. Implementing multiple security protocols and combining them strengthens the network. Even if one control is breached there are still additional mechanisms in place to provide security. The defense in depth principle is reflected in the merged network through the use of Demilitarized Zones (DMZs) and network segmentation. Firewalls are inserted throughout the network to create DMZs, creating an additional layer of security in the network. The firewalls are placed at network entry points and separate the internal and external networks meaning that all network traffic is closely monitored. Also, if an attacker gains access to one part of the

network, the remaining firewalls are in place to defend the rest of the network. Furthermore, it's easier to contain attacks and address security incidents quickly when the network is segmented.

Principle #2 — Least privilege. In addition to incorporating the defense in depth principle into the network, it was important to add aspects of least privilege as well. Before the network was merged, every employee at Company A had full access privileges to all systems except for payroll. Least privilege means that employees should only be granted access to the resources necessary for their respective job roles. To implement least privilege in the new network, open ports used for remote desktop were closed and Role Based Access Control (RBAC) is used so users have preset permissions based on their roles in the company (Chapple et al., 2021). Integrating defense in depth and least privilege principles allows the IT department to minimize the impact on the network when user accounts or network components are compromised. It also reduces the risk of data breaches and unauthorized access to the network.

H. Describe two secure hardware and/or software components integrated into the proposed network topology and how *each* component will address the security needs of the merged organization.

Various components were incorporated into the merged network to create a protected network environment. Two of the components implemented are Cisco Catalyst 2960X 48 Port Switches and Windows 10 with built-in security features.

#1. Hardware — Cisco Catalyst 2960X 48 Port Switches. Many of the hardware components at both companies were outdated and would be security threats in the merged network. The decision to invest in new switches is important for several reasons. The newly purchased switches will receive ongoing support from Cisco

including software patches that will keep the switches up to date. When addressing network security, using hardware with updated firmware is critical. The switches currently in place don't meet the security needs of the merged organization that will handle sensitive financial services and insurance data. The upgraded switches are also scalable and futureproof to keep up with advancements in security technology. The newer Cisco switches are equipped to handle encrypted remote management protocols such as Secure Shell (SSH) and Hypertext Transfer Protocol Secure (HTTPS). They also are equipped with Access Control List (ACLs) so IT administrators can restrict access to specific ports and dictate what traffic is allowed or denied on the network.

#2. Software – Windows 10 with BitLocker Drive Encryption and Windows Defender Antivirus. As previously stated, the decision to purchase Windows 10 licenses was based on the many security features that are already built into the operating system at no additional cost. One of the features that will help secure the merged network is BitLocker Drive Encryption. BitLocker makes user devices more secure because it encrypts the data on the devices' drives. In the case that a device is lost or stolen, the data on the drives cannot be accessed without the encryption key.

While the network firewalls will filter out malicious network traffic, Windows Defender Antivirus will scan devices for malware and automatically update to the newest firmware. Windows Defender is another way for us to monitor the network for unauthorized activities and keep it secure. Since the antivirus software runs in real-time in the background on the device, it is constantly monitoring the system and blocking threats with no intervention necessary from an IT administrator.

I. Explain how the proposed network topology diagram for the

merged organization addresses security safeguards based on a regulatory compliance requirement.

The proposed network topology diagram for the merged organization incorporates various security safeguards to address regulatory compliance requirements, including the Health Insurance Portability and Accountability Act (HIPAA), and the Gramm-Leach-Bliley Act (GLBA).

Regulation #1 — Health Insurance Portability and Accountability Act (HIPAA). As an insurance underwriting company that underwrites different types of insurance, Company A falls under the umbrella of business entities that have to abide by HIPAA. HIPAA requires covered entities to protect the privacy and security of patients' health information. In the merged network, DMZs and other access management protocols are in place to ensure that sensitive data is only available to users who need access. This fulfills HIPAA's information access management requirement, which states an organization must use technical mechanisms to protect the integrity and confidentiality of patient information.

Regulation #2 — Gramm-Leach-Bliley Act (GLBA). The GLBA has requirements that pertain directly to financial organizations such as Company B. The regulation requires financial institutions such as banks and investment firms to protect customer data. The GLBA has several requirements, including a written information security program (WISP) which this proposal serves as. After a risk assessment has been completed to identify the vulnerabilities that exist in protecting customers' private information, a company is legally required to mitigate them. The components and changes being recommended for the merged network such as enforcing password changes and purchasing updated routers establishes our compliance with the GLBA.

Similar to HIPAA, the GLBA requires that access controls be in place which will be accomplished through the use of VPN and Role Based Access Control (RBAC).

J. Explain a security threat and one potential network problem that would become a risk as part of the implementation of the proposed network topology diagram, including why *each* would become a risk.

Although a considerable effort has been made in this proposal to secure the network and prevent any cyberattacks in the future, vulnerabilities and threats will always exist. It is the responsibility of IT administrators to continuously monitor the network and take action to patch vulnerabilities and protect proprietary data.

Security Vulnerability — Cisco 4331 Router Firewall and VPN. In the merged network, the firewalls and VPNs are provided by the software on the routers as opposed to using third-party services. Even though Cisco is a reputable company known for its secure networking devices, the use of the router for firewalls and VPNs can be a vulnerability. Because the firewall and VPN services are provided by the router, an attack on the device means those services are more susceptible to being compromised in an attack. If the routers and firewalls were attacked, unauthorized users could gain access to the network creating a data breach.

Potential Network Problem — Network Congestion. As the organizations merge and new components are added to the network, network traffic will increase substantially. After the merger, users may experience reduced network speeds, decreased performance, and service disruptions due to more devices being in the network. This can lead to data loss through dropped packets and lower productivity rates as users struggle with connectivity.

J1. Explain how the security threat and potential network problem should *each* be managed or mitigated as part of the implementation of the proposed topology diagram.

Mitigating Security Vulnerability — Cisco 4331 Router Firewall and VPN. To protect the security and integrity of the router, it's important to apply a layered security approach. Two-factor authentication (2FA) can be applied to prevent unauthorized users from accessing its settings and making changes in an attack. 2FA will also be a safeguard against human error ensuring that individuals who lack the knowledge base to properly configure the router don't have access to it. To further mitigate the associated risks, the firewall should be regularly monitored by the IT department for suspicious activity and checked for firmware updates often.

Mitigating Network Problem — Network Congestion. Once the organization has merged, network performance should be monitored regularly and strong incident response policies need to be in place. This involves reviewing the design of the network as needed to handle increased traffic, and monitoring network performance and bandwidth utilization. As the organization continues to grow, network requirements may change and network upgrades or bandwidth expansion should be considered. By addressing vulnerabilities and proactively managing network problems, the organization can maintain a secure and efficient environment for the foreseeable future.

In today's world where technology plays a pivotal role in driving growth and maintaining operational efficiency, upgrading the network infrastructure is a necessary investment. While the current networks at both companies have been functional, they won't meet the demands of the merged network and will become incapacitated as the

organization continues to grow. An upgraded network will provide tremendous benefits and a competitive edge in the insurance and financial services industry. Redesigning the combined network according to this proposal will ensure that it has improved performance, security, scalability, and compliance with information security regulations.

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